

SYLLABUS :-

AR23201: ARCHITECTURAL DESIGN I Credits: 5 L-T-P: 2-0-4 Prerequisite: Basic Design I

II Course Overview: This sessional is the first fundamental introduction to Architectural Design Process. The students shall be introduced to various theories and approaches towards architectural design process through two design problems. The first, being an individual Residence Design, which shall have greater focus on spatial and functional requirements of the users specified. The second, a competitive group work, time constrained, design problem with greater focus on Architectural aesthetics and form development. The Students shall be introduced to all constituents in the architectural design process, spanning from site analysis, space programming, conceptual design, detailed design development and presentation techniques. The course shall encourage in students, independent creative thinking and analytical approach towards design development while familiarizing them with architectural drawings preparation techniques and presentation skills. **Learning Objectives:**

1. Understand various theories and approaches towards architectural design process and its constituent parts.
2. Learn to interpret user's or client's spatial requirements and preferences and analyse its impact on design development.
3. Understand the process and constituent parts of site analysis, area programming, adjacency matrix, bubble diagram, site zoning and conceptual or schematic design development.
4. Get acquainted with the techniques of final design preparation (plans, elevations, sections, views etc.) while exploring, architectural aesthetics in façade development and sheet presentation techniques.
5. Re-examine design skills learnt in a competitive, time constraint, unconventional design problem to enhance creative performance in design development.

Course Curriculum:

Module 1 : Introduction to Architectural Design Process Introduce to the students the various theories and approaches towards architectural design process and its constituent parts. Introduction to the design problem – Residence design with specific client requirements in a specific city.

Module 2 : Site Analysis, Area and Space Programming Building byelaws, site access, climatological considerations (sun-path, wind direction, climatic zone, orientation etc.) associated with site. Area programming of all spaces required as per design brief taking references from architectural and anthropometrical standards. Introduction to and preparation of adjacency/proximity/criteria matrix.

Module 3 : Conceptual Design Development Introduction to and preparation of bubble diagrams based on space programming. Preparation Site Zoning Plan (based on climatological, noise and privacy considerations)

Module 4 : Final Design Development Preparation in order of - Single line plan, Double line plan (Site plan, All floor plans, Terrace floor plan) with dimensioning furniture layout, Elevations, Sections with proper dimensioning and levels and appropriate 3 D views. Introduction to building aesthetics and façade design alongside sheet presentation techniques and compositions.

Module 5 : Introduction to Team work Dynamics in Architectural

Design Process An additional team work based design problem to be pitched in the form of a Design Challenge on an unconventional site condition and with time constraint. Familiarization to competitive design development with focus on aesthetics/form and design innovations while discovering team work dynamics and time management. Suggested projects: Residence Design as main project, community hall, Anganwadi, café, rural centre, panchayat office etc. with innovative site constraints as group project. Reading List: 1. Callender J. Hancock, Time Saver Standards – A Handbook of Architectural Design. 2. Snyder, J.C. and Catanese, A.J., Introduction to Architecture. 3. Porter, T., Colour for Architecture. 4. Meiss, V. and Pierre, Elements of Architecture: From Form to Place. 5. Time saver standards 6. Neuferts 7. Pen and ink rendering